

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \xrightarrow{\text{inverse}} \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

1. swap the two diagonal numbers
2. flip the sign of the other two
3. divide everything by $ad - bc$

$$\begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix}^{-1} = \frac{1}{2 \times 13 - 5 \times 5} \begin{bmatrix} 13 & -5 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 13 & -5 \\ -5 & 2 \end{bmatrix}$$

check: $\begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix} \begin{bmatrix} 13 & -5 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 26 - 25 & -10 + 10 \\ 65 - 65 & -25 + 26 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, the identity